

# IRFAN KHAN

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## Education

**National Institute of Technology, Jalandhar**

2023 – 2027

*B.Tech in Electronics and Communication Engineering (CGPA: 7.57)*

Jalandhar, Punjab

**Kendriya Vidyalaya No. 1**

2022 – 2023

*Higher Secondary Education (Class XII) - 91.2%*

Jalandhar Cantt

## Technical Skills

- **Languages & Hardware:** Verilog HDL, C/C++, Python, MATLAB
- **Tools & EDA:** Vivado, GTKWave, Arduino IDE, EDA Playground, Tinkercad, Wokwi, KiCad
- **Core Areas:** RTL Design, FPGA Implementation, CMOS Design, Digital Electronics, IoT

## Projects

**RISC-V Single Cycle 32-bit Processor** 🌀 Verilog, RTL Design, Icarus Verilog, GTKWave

- Architected a **32-bit single-cycle RISC-V processor** implementing the **RV32I ISA** using modular RTL design
- Deployed datapath including **Program Counter, Register File (32×32), ALU, Control Unit, and Memory Interfaces**
- Executed arithmetic, memory, and branch instructions with **single-cycle latency**
- Performed functional verification using a **custom Verilog testbench** and waveform debugging in GTKWave
- Optimized datapath and control logic for **reduced combinational delay and efficient hardware utilization**

**Asynchronous FIFO Design** 🌀 Verilog, CDC Design, GTKWave

- Created a **parameterized asynchronous FIFO** supporting **dual clock domains** with reliable CDC using **Gray-code synchronization**
- Implemented **two-flop synchronizers** to mitigate metastability during clock domain crossing
- Built robust **FULL and EMPTY flag logic** using an **N+1 pointer scheme** ensuring no overflow/underflow
- Verified functionality under asynchronous conditions using a **comprehensive testbench with randomized scenarios**
- Ensured timing-safe CDC design aligned with **industry RTL design practices**

**CMOS-Based Neural Network Accelerator** Verilog, FPGA, Vivado 2025 – Present

- Engineered a **hardware accelerator** for real-time ECG signal processing and heart rate detection
- Designed optimized datapath for **low-latency and power-efficient computation** using CMOS-based logic techniques
- Synthesised and implemented the design on FPGA using Vivado for efficient resource utilisation
- Performed simulation and validation of neural computation blocks for accurate signal classification
- Explored hardware-software co-design concepts for embedded neural inference systems

## Experience & Leadership

**4-Time SSB Candidate** — *Indian Army Selection*

- Cleared multiple stages of **Service Selection Board (SSB)** selection process, demonstrating leadership, decision-making, and psychological resilience

**SoECE Club Member** — *NIT Jalandhar* 2023 – Present

- Active member contributing to technical workshops and hardware activities
- Organized SoECE Cricket Tournament for 100+ students

**NCC Cadet** — *National Cadet Corps* 2021 – 2023

- Mentored junior cadets and developed leadership and discipline skills

## Achievements

- **National Level Basketball Player:** Represented at the national level, demonstrating teamwork and high-performance determination
- **Academic Excellence:** Secured 91.2% in Class XII and currently maintaining a CGPA of 7.57 at NIT Jalandhar